Project Report

Eco Treasurer: Turn Waste Into Wander

Course code: CSE 428

Course Title: Human Computer Interaction

Section: 01

Group Name: Neosis

Submitted by

Name	ID
Tithi Paul	2021-2-60-057
Nirzona Binta Badal	2021-2-60-051
Marzan Billah Shefa	2018-3-60-112
Adnan Mahmud	2018-3-60-023

Submitted To

Yasin Sazid
Lecturer

Department of Computer Science & Engineering

East West University

Date of Submission: 22 May, 2025

Declaration

We, **Nirzona Binta Badal, Tithi Paul, Marzan Billah Shefa and Adnan Mahmud** hereby declare that the work presented in this project report, titled **"Eco Treasurer: Turn Waste Into Wander"**, is the result of our independent investigation and development, conducted under the supervision of **Yasin Sazid**, Lecturer, Department of Computer Science & Engineering, East West University, Dhaka, Bangladesh. We affirm that no part of this project has been or is being submitted elsewhere for the award of any degree or diploma, except for publication purposes.

Countersigned

Yasin Sazid

....

Yasin Sazid
Supervisor

Signatures

....

Nirzona Binta Badal(2021-2-60-051)

Tithi Paul

....

Tithi Paul(2021-2-60-057)

Shefa

....

Marzan Billah Shefa (2018-3-60-112)

Adnan

....

Adnan Mahmud (2018-3-60-023)

Table of Contents

List of Figures

List of Tables

Chapter 1: Introduction

1.1 Project Theme

The subject of this project is sustainability through upcycling, where the focus is placed on promoting eco-friendly living by transforming waste into valuable resources. During a period when nature is degrading and additional waste is being produced, this project has been set to address demands for conscientious consumption and creative reuse, especially among the urban population.

1.2 Project Overview

EcoTreasure is a mobile application that encourages sustainable living by enabling users to find, create, and share upcycling products and ideas. The app offers video tutorials on do-it-yourself projects from recyclable materials, a community-driven marketplace for promoting and selling handmade products, and utilities to locate local eco-stores and sustainability events. It is also loaded with educational content in the form of articles and journals for generating awareness and inducing behavioral change. The system utilizes cutting-edge interaction design merged with real-world eco-solutions to enable users to make effective and informed lifestyle decisions.

1.3 Objectives and Scope

- Objectives:
    1. Encouraging sustainable practices through upcycling and DIY activities.
    2. Empowering users through knowledge and tools for the creative re-use of waste.
    3. Fostering a community of users who share a common passion for sustainability.
    4. Building a local marketplace for eco-friendly handmade products.
    5. Connecting users to local environmental resources such as green shops and green events.

- Scope:

  This application is aimed at eco-aware people especially young adults and students with an interest in technology and sustainable practices. There is a module for user registration, browsing content (videos, journals), marketplace activities (buying and selling), messaging, and profile personalization. The first prototype is designed for an urban Bangladeshi setting but can be scaled and adapted to other places.

1.4 Stakeholders

- **Primary Users:** Individuals (students, hobbyists, artisans) concerned with eco-friendly living, crafts, and upcycling.
- **Eco Entrepreneurs:** Micro to small-scale businesses and local merchants dealing in sustainable products and services.
- **Community Members:** Creators, advocates, and educators focusing on sustainable development who volunteer for tutorials, articles, and events.
- **Administrators:** The app developers and content editors are responsible for the technical aspects of the app, such as function, user interaction, content moderation, and a healthy platform.
- **Environmental NGOs/Organizations:** Strategic partners who would be interested in sponsoring campaigns, workshops, and collaborative content development.

Chapter 2. Low Fidelity Sketching

We began designing EcoTreasure: Turn Waste into Wonder! With low-fidelity sketching to visualize basic ideas and iterate quickly. Low-fidelity (lo-fi) sketches are essential early user experience design tools. They allow the quick investigation of ideas without being constrained by high detail levels or finished appearances. By sketching various themes, we were able to try out alternative layouts, flows, and features that supported the app's goal: promoting sustainability through creative reuse.

Overview of Themes

1st Theme: This theme focused on the marketplace and user-driven creativity. Sketches included upcycled product listings, material-based filters, and sections for DIY ideas. We also visualized a daily eco-tip feature to foster user engagement through bite-sized sustainability knowledge.

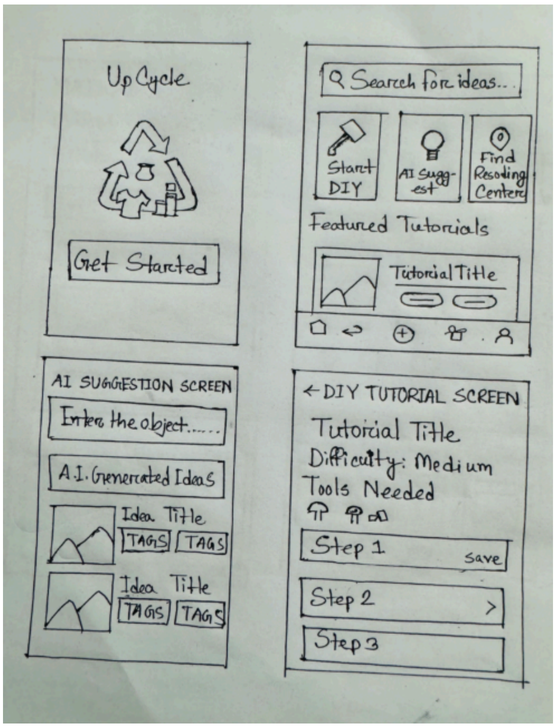


Figure 1: Lo-fiTheme 1 Part 1

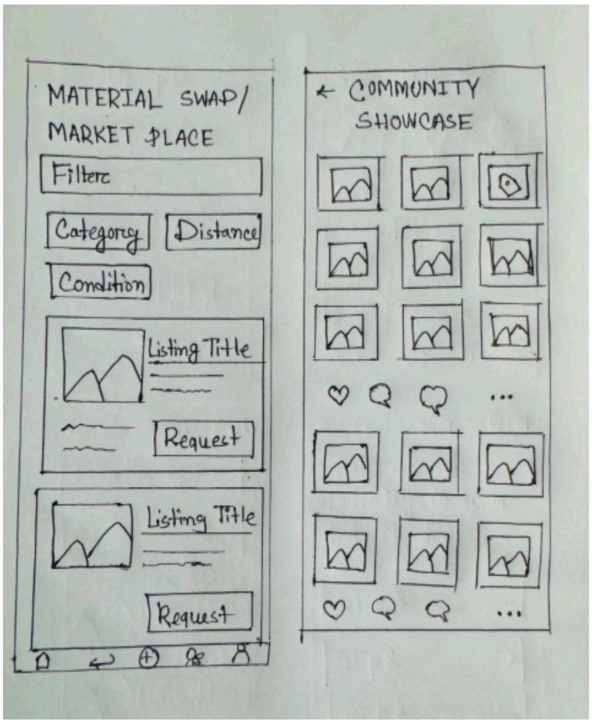


Figure 1.1: Lo-fi Theme 1 Part 2

2nd Theme: In this theme we explored a personalized dashboard where users could log eco-friendly actions, set sustainability goals, and receive activity recommendations. Elements such as progress bars, habit trackers, and eco-point systems were introduced to gamify environmental actions.

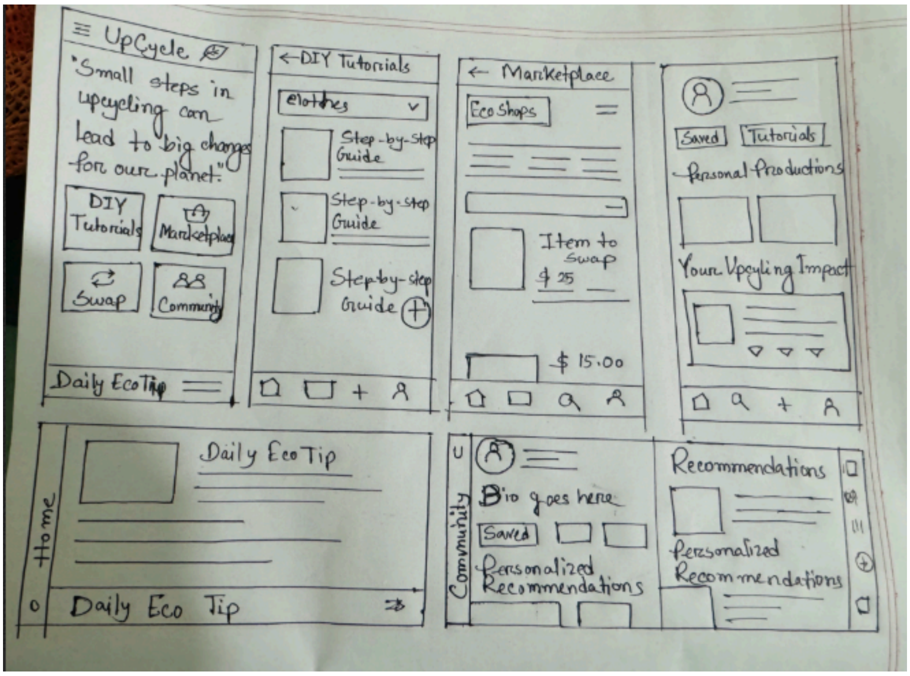


Figure 2: Lo-fi Theme 2

3rd Theme: This theme features designs for tutorial modules and community content. These included tutorials, user-generated posts, and local eco-events. The sketches emphasized learning through shared experiences and fostering a supportive, active community.

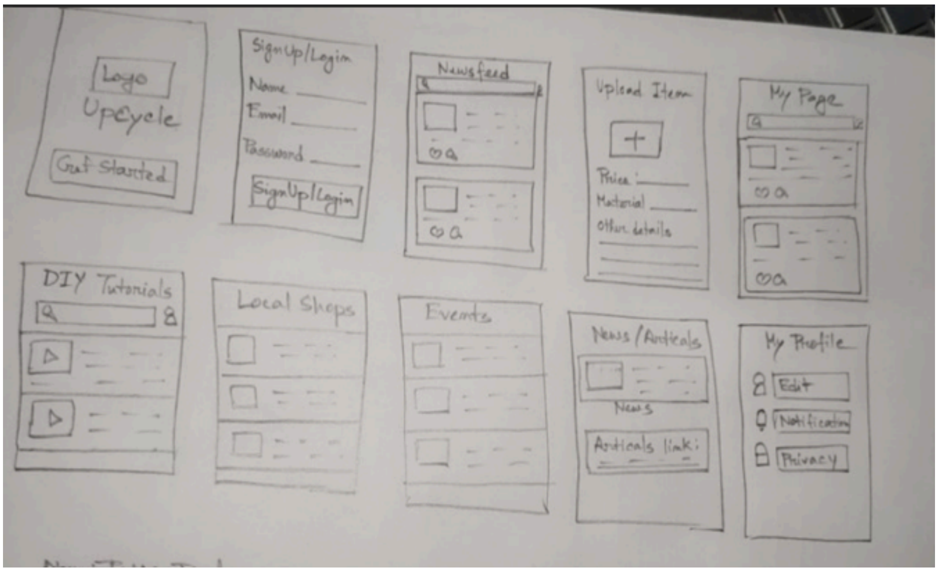


Figure 3: Lo-fi Theme 3

4th Theme: The fourth theme sketched out user profiles, eco-badges, and visual trackers to display environmental contributions. This helped reinforce personal accountability and allowed users to monitor their sustainable practices over time.

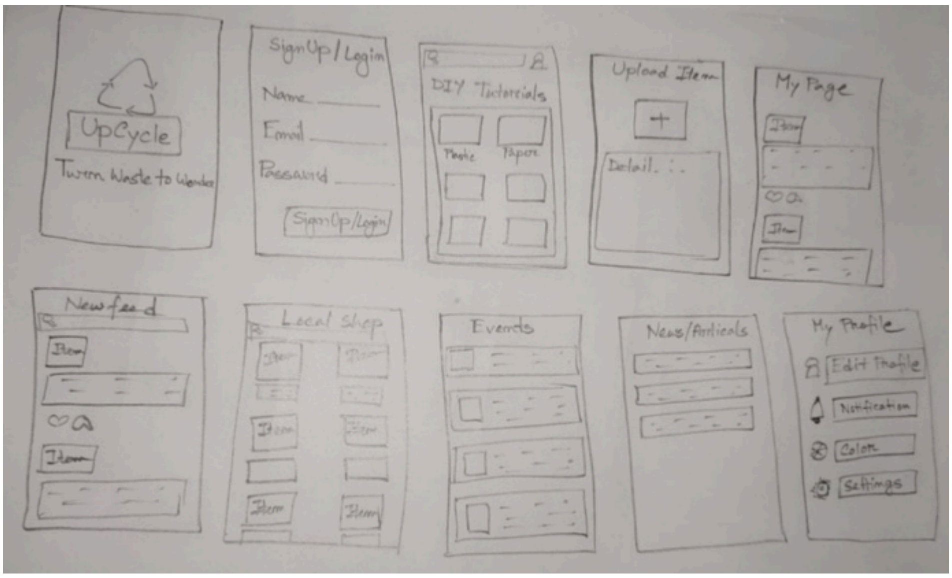


Figure 4: Lo-fi Theme 4

Chapter 3. Background Study

3.1 Related Work

In order to lay the groundwork for our project, we analyzed various academic papers concerning sustainability, upcycling, and their digital applications fostering environmentally-friendly actions. These studies analyzed in detail the role of technology in supporting sustainable living, modifying user behavior, and addressing the challenges associated with upcycling. The table below captures the salient points of each research work we reviewed.

Table 1: Related works

Paper Title	Methodology	Data Sources	Key Result	Future Opportunities
Mobile app-aided design thinking approach to promote upcycling in Singapore	Applied a mobile app prototype based on design thinking principles to engage users in sustainable behavior via upcycling tasks.	Survey responses and behavioral data from users in Singapore using the mobile app.	Demonstrated improved environmental awareness and engagement in upcycling activities through digital interventions.	Scaling the study across diverse demographics, integrating gamification and AI-based recommendations.
Upcycling Mobile App as a Way to Raise Awareness About Sustainable Living and Decrease Fashion Waste	Developed and tested an upcycling app targeting fashion waste, focusing on awareness campaigns and DIY content.	Fashion waste data and user engagement analytics from the app.	Users reported increased motivation to upcycle and reduce fast fashion dependency.	Integrating AI for fashion recognition, community marketplace, and better content personalization.

Upcycling Systems Design: Developing a Methodology through Design	Proposed a methodological framework for upcycling systems design using iterative design and stakeholder feedback.	Case studies and iterative prototypes tested with users.	Helped define systemic design tools for sustainable product development.	Applying the framework to commercial product lines and larger industrial systems.
Implementing a Cutting-Edge Recycling Web Application in Tha Pho Sub-district	Developed a recycling web app to improve collection logistics and participation.	Local waste data and stakeholder interviews in Tha Pho, Thailand.	Increased recycling rates and stronger collaboration between residents and industry.	Expansion to other districts, integrating AI-based sorting assistance.
Motives, Drivers and Barriers to Urban Upcycling (Netherlands)	Qualitative study using interviews to understand motivations in furniture upcycling.	Interviews and case studies from Dutch upcyclers.	Identified strong environmental and aesthetic drivers, with lack of time and skills as barriers.	Design policy interventions and educational programs to reduce skill and time barriers.
Upcycling Junk Art and Craft Module to Nurture Children's Creativity	Designed a junk art module for children to promote creativity via upcycled materials.	Classroom implementations and pre/post creativity assessments.	Boosted creativity scores and increased awareness of sustainability.	Incorporate modules in national curriculum and test long-term behavioral change.
The Sustainability Potential of Upcycling	Evaluated upcycling's environmental impact using	Comparative LCA data across materials.	Upcycling significantly reduced environmental load versus	Apply standardized upcycling metrics and promote policies for adoption.

	life cycle assessments.		recycling or landfill.	
Critical Advances and Future Opportunities in Upcycling Commodity Polymers	Literature review on chemical and mechanical upcycling methods for polymers.	Existing research and case studies from polymer upcycling.	Highlighted technological gaps and promising approaches.	Push for scalable, closed-loop systems and economic feasibility studies.
Sustainable Design Strategy via Upcycling Waste Wood + Digital Marketplace for Micro-Crafters	Merged sustainable design using wood waste with digital marketing tools.	User data and marketplace interaction stats.	Enabled micro-entrepreneurs to profitably sell upcycled wood crafts.	Expand to global artisan networks and add AI-driven pricing/recommendation features.

3.2 Summary

From the reviewed literature, we identified recurring themes such as the effectiveness of mobile apps in evoking environmental awareness, the gamification power to engage users, and social participation. The majority of studies highlighted the effectiveness of design thinking and user-centered approaches in apps focusing on sustainability. These findings guided our own design decisions in EcoTreasure, such as incorporating DIY instructions, a marketplace for the community, and motivational features like eco-points and achievements. In addition, the possibilities mentioned in these reports—i.e., integrating AI, scalability to large user bases, and enabling micro-entrepreneurs—shaped the future direction and scalability of our project.

Chapter 4. User Research

4.1 Research Questions Development

<u>**Research Question 1**</u>

How do users currently engage in sustainable or eco-friendly practices in their daily lives?

- **Motivation:**

  Understanding users' existing habits and behaviors related to sustainability helps us design features that complement or enhance their current routines rather than disrupt them.

- **Participant Information to Collect:**

  - ➔ Habits regarding recycling or reusing
  - ➔ Attitudes toward sustainability and eco-friendly living

- **Design Insight Contribution:**

  Insights will help us prioritize features like upcycling tutorials, habit trackers, or daily green tips based on users' existing levels of eco-engagement.

<u>**Research Question 2**</u>

What challenges do users face when trying to follow sustainable practices, and what support do they expect from an app like EcoTreasure?

- **Motivation:**
  Identifying barriers can help us design solutions that address specific user pain points, such as lack of knowledge, motivation, or time.

- **Participant Information to Collect:**

- ➔ Challenges or difficulties with sustainability
- ➔ Expectations from an app that promotes eco-friendly actions

- **Design Insight Contribution:**
  This will guide us to design helpful, motivating, and supportive app features, such as simple tutorials, a reward system, or a supportive community space.

4.2 Interview Questionnaire

Supporting Research Question 1

1. Can you describe any eco-friendly or sustainable habits you currently follow?
2. How often do you recycle, reuse, or upcycle items at home?
3. What motivates you to follow sustainable practices?
4. What kind of content or information would help you live a more eco-friendly life?

Supporting Research Question 2

5. What are the biggest challenges you face when trying to be more sustainable?
6. Have you ever looked for apps or digital tools to support your sustainable habits? If yes, what did you like/dislike?
7. What features would you expect in an app that promotes sustainability?
8. How likely are you to use an app that offers DIY tutorials or community challenges related to sustainability?
9. How important is visual guidance (photos/videos) to you when trying DIY or upcycling ideas?

4.3 Alternate or Complementary Research Method

To complement our interview questions, we created an **online survey** using Google Forms. This format allowed us to collect a broader range of data efficiently from a larger group of participants. Many interview-style questions were adapted into multiple-choice and rating-scale formats for easy analysis.

The survey covered:

- Frequency of eco-friendly actions
- Personal motivations and barriers
- Feature preferences for sustainability-focused apps
- Demographic details to help understand context

4.4 Data Collection

Survey Summary

We collected responses from **30 participants** through our Google Form. Participants were mostly university students with varying levels of sustainability engagement.

Key Findings from 3 Interview Participants

- **Participant 1**: Recycles plastic but struggles with finding DIY ideas. I Want simple tutorials.
- **Participant 2**: Interested in sustainability but lacks motivation. Would like a reward-based system.
- **Participant 3**: Loves DIY but doesn't know where to start. Prefers visual content (videos/images).

Design Insights Derived

The user study yielded some of the most influential design findings for the direction of the 'EcoTreasure' prototype. Most of the users indicated a demand for simple-to-follow, visually based DIY content, which implies having simple-to-follow visual instructions and labels like "beginner" to facilitate usability. Motivation was also a concern, which could be resolved by incorporating gamification features like point systems, achievements, or badges to make it more engaging for the users. A frequent barrier was unfamiliarity with upcycling or sustainability, which means that there is scope for item-type segmented tutorials and bespoke eco-tips. As an encouraging finding, most participants were already sustainability-interested, making it possible to add community-based aspects like challenges or idea-sharing forums. Finally, the users stressed the significance of simplicity and aesthetics, pointing out the necessity of a simple, clean, and visually appealing interface that would be easy to navigate, particularly for first-time users.

Chapter 5. High Fidelity Prototyping

5.1 Figma Prototype URL

Our high-fidelity prototype was designed using **Figma**, a collaborative interface design tool. The prototype includes all the main features of the EcoTreasure app, such as the Welcome screen, Login/Signup, Home page, DIY Tutorials, Marketplace, Local Shops, Events, Journal, Profile, Messaging, and Settings pages.

5.2 Style Guide

To ensure a consistent and user-friendly interface, we followed a simple and modern style guide throughout the app:

- **Typography**:

  EcoTreasure uses three primary font styles to ensure a playful yet readable interface:

  1. **H1 – Irish Grover**
     - Style: Regular
     - Size: 36px
     - Use: App titles and major headings
  2. **H2 – Instrument Sans**
     - Style: Regular – 24px
     - Substyles:
       - **T1**: Bold – 20px
       - **T2**: Italic – 16px
     - Use: Section headers, subtitles, and emphasis

3. **H3 – Comfortaa**
    - Style: SemiBold – 16px
    - Substyle:
        - **T1**: Medium – 16px
    - Use: Body text and labels

Note: Font sizes may be adjusted slightly based on layout needs or screen sizes for better responsiveness and legibility.

- **Color Palette**:

The following colors are used consistently throughout the EcoTreasure app for a clean and eco-friendly look:

- **Deep Green** – #24362E
- **Light Green** – #689C85
- **Parrot Green** – #97C816
- **White** – #FFFFFF
- **Gray** – #B9B9B9
- **Black** – #000000

These colors support a cohesive, user-friendly, and visually appealing design across all app components.

- **Buttons**:
    - Rounded corners
    - Solid fill for main actions

- Outlined or text buttons for secondary actions

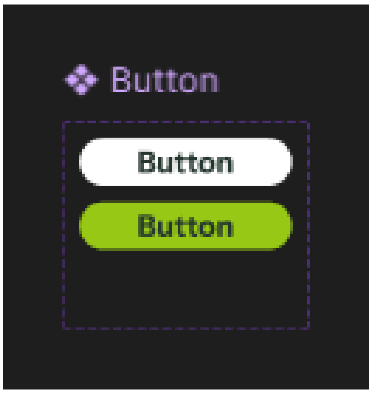


Figure 5: Button

- **Icons**:

- Simple, line-style icons for better clarity

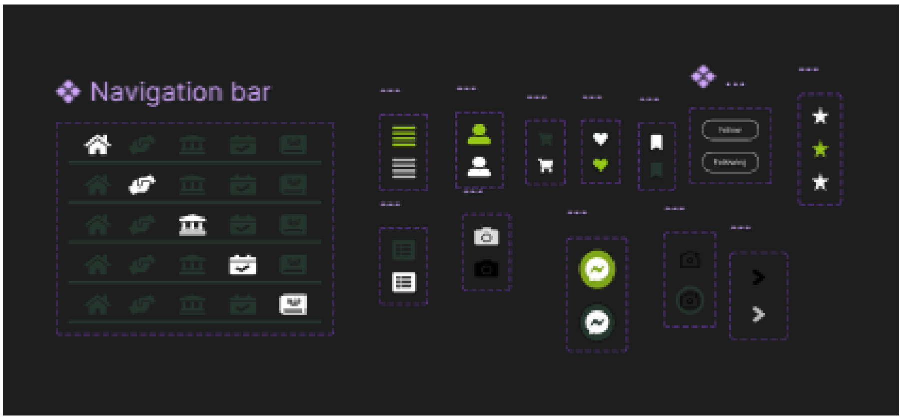


Figure 6: Navbar

- **Components:**
  - Text Fields: For titles, messages, and inputs with clear labels and placeholders.
  - Search Bars: With icons for easy navigation and filtering.
  - Marketplace Cards or Home or Profile contents: Show image, title, and info for listed items.

- Message Input & Keyboard: For user messaging and emoji input.
- Gallery Selector: Upload or select images easily.
- Feeling Tags: Let users express emotions in posts.
- Description Boxes: Input areas with soft, readable design.
- Loading Bar: Visual feedback during upload or transitions.
- Video Cards: Tutorial previews with thumbnails.
- Grouped Buttons: For settings, theme, logout, etc.
- Message Previews: Quick view of recent chats.

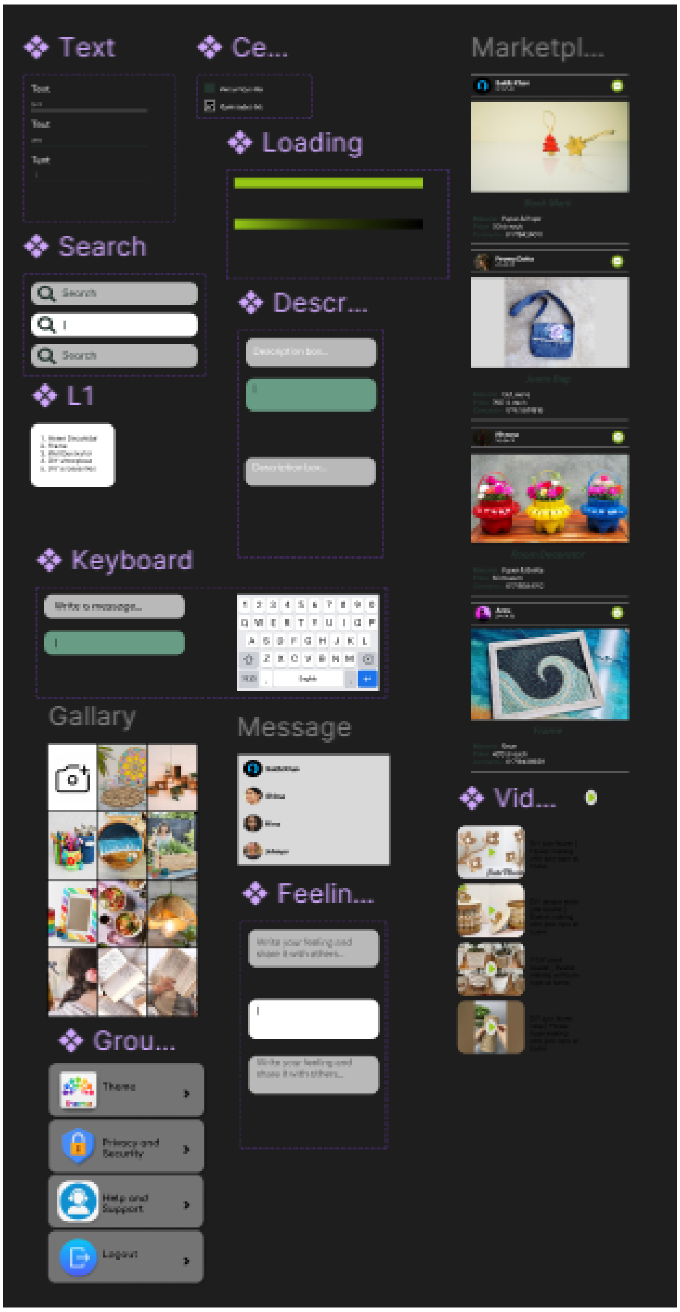


Figure 7: Components

- **Imagery**:
  - Clean and relevant visuals
  - Icons and illustrations that reflect sustainability, recycling, and creativity

This style guide helps maintain a clean, eco-friendly, and approachable look across all app screens.

5.3 Prototype Screens (Screenshots)

Below are selected screenshots from our high-fidelity Figma prototype that visually demonstrate the user interface and main features of the **EcoTreasure** app. These images showcase the actual design implementation of our style guide, layout decisions, and user flow.

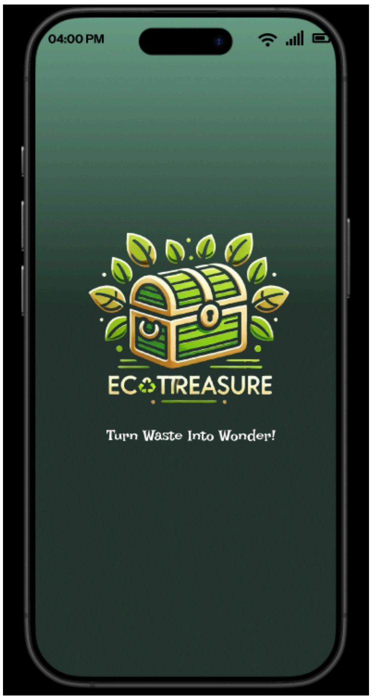


Figure 8: Splash Screen Figure 9: Loading Screen Figure 10: Welcome page

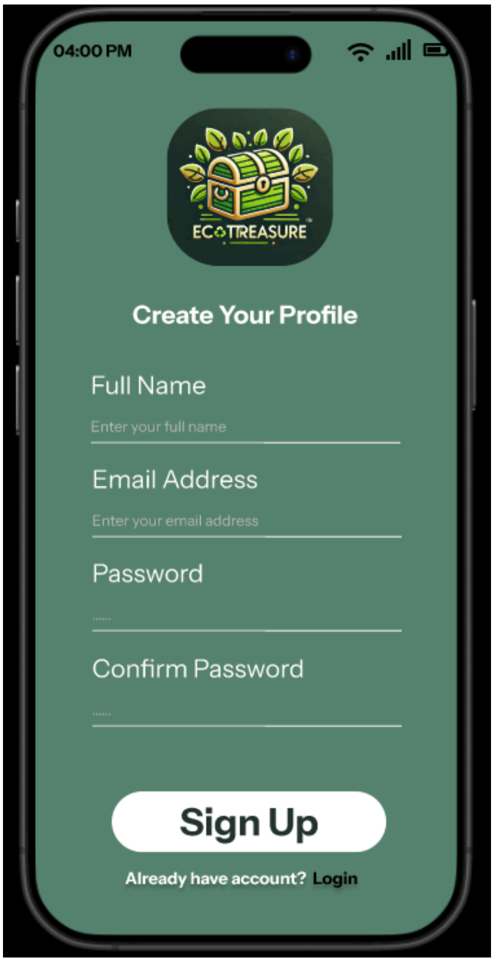


Figure 11: Sign Up Page

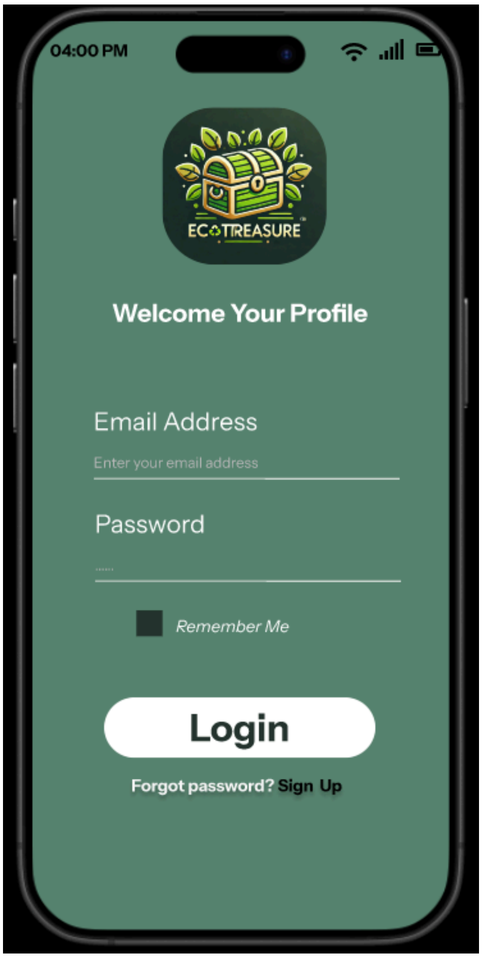


Figure 12: Login Page

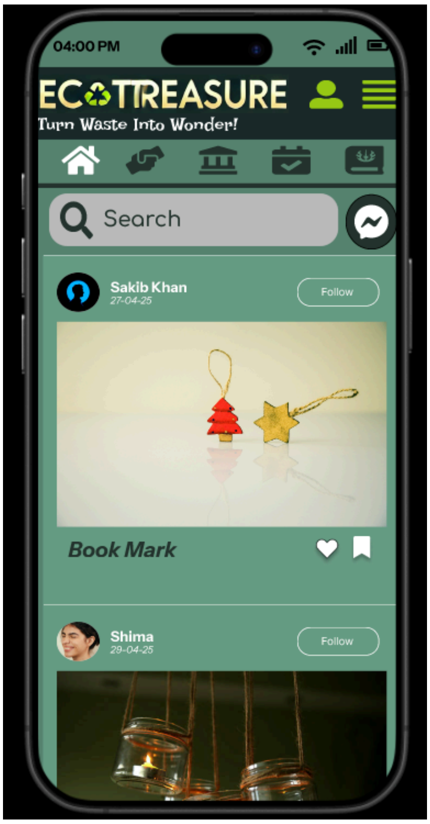


Figure 13: Home Page

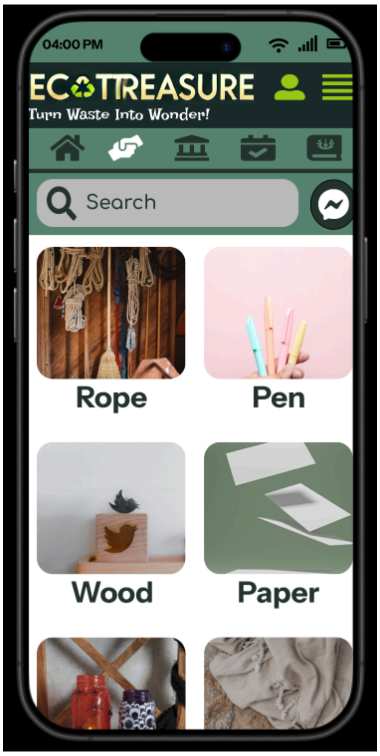


Figure 14:DIY Content Page

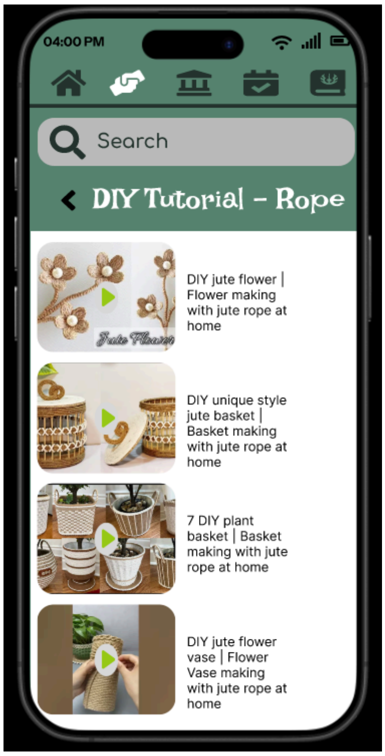
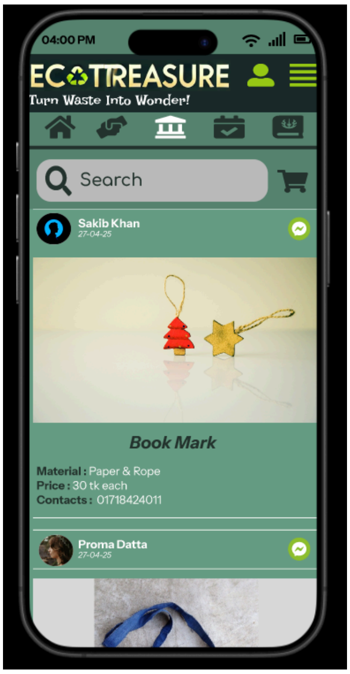


Figure 15:DIY Video Tutorials Page



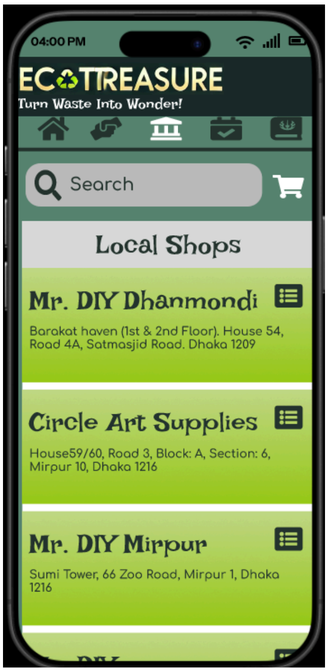
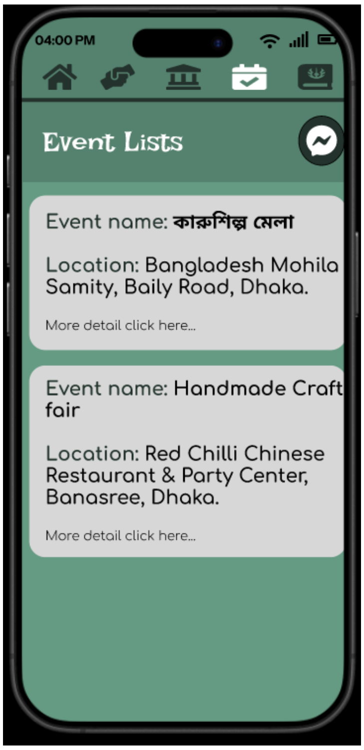


Figure 16: Market Place Page

Figure 17: Local Shop Page



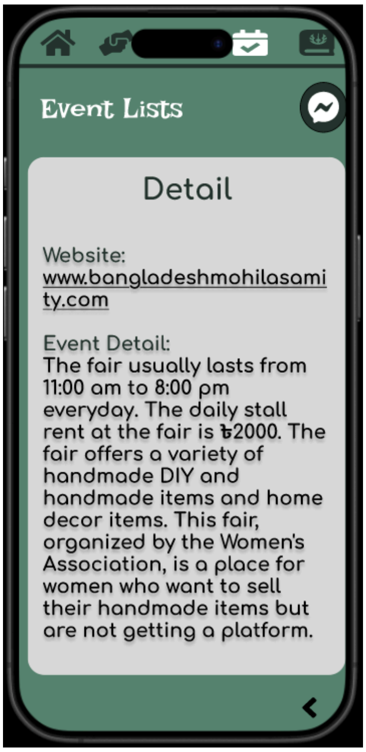


Figure 18: Event List Page

Figure 19: Event Details Page

Figure 20: Journal Page

Figure 21: Journal Details Page

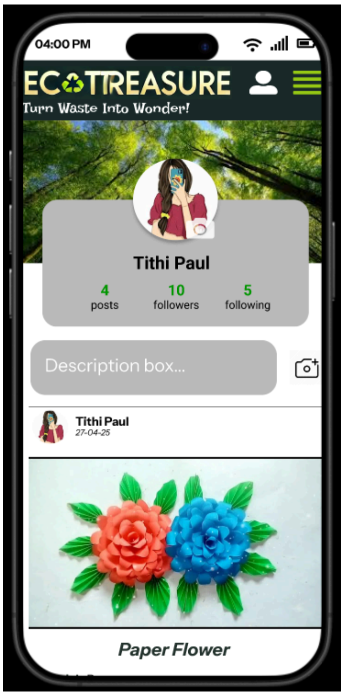


Figure 22: Profile Page

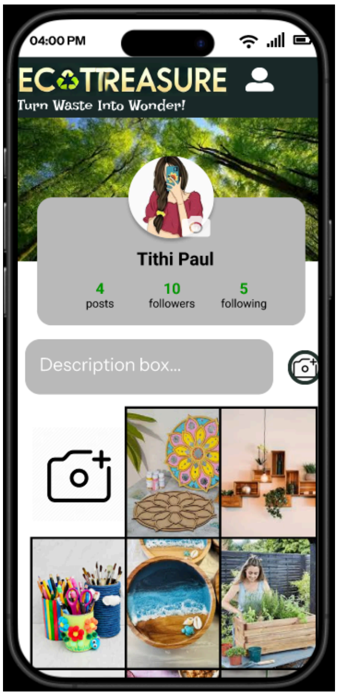


Figure 23: Add picture page

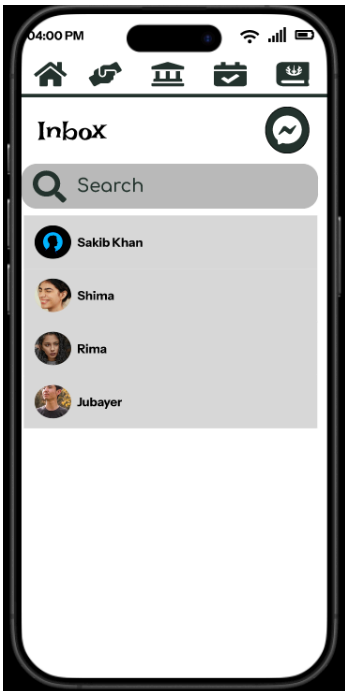


Figure 24: Massage Page

Figure 25: Inbox picture

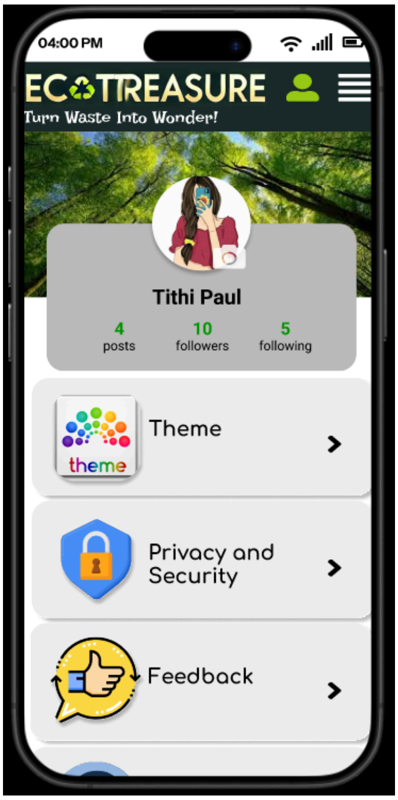


Figure 26: Settings Page

Figure 27: Log Out Page

Chapter 6: Interface Evaluation

6.1 Usability Evaluation Using Nielsen's 10 Heuristics

Table 2: Usability evaluation

Heuristic	Usability Problem	Severity	Proposed Solution
1. Visibility of system status	After uploading a DIY project, there is no immediate confirmation or progress indication.	Minor	Add a loading indicator and a success message like "Project uploaded successfully."
2. Match between the system and the real world	Some users were confused by the "My EcoBoard" label; it's unclear whether this refers to saved or favorited items.	Minor	Rename it to "Saved Projects" or "Favorites" to better align with user expectations.
3. User control and freedom	No 'Undo' option for deleting a project post. Once deleted, the content is permanently gone.	Major	Implement a "Trash" or "Recently Deleted" feature that allows restoration within 7 days.
4. Consistency and standards	The bottom navigation bar uses both icons and labels	Minor	Ensure all tabs use both icons and text for clarity and

	inconsistently (some tabs have only icons).		consistency.
5. Error prevention	Users can accidentally submit a blank journal post without content validation.	Major	Add form validation to check for title and body content before allowing submission.
6. Recognition rather than recall	Users must remember the steps to publish a tutorial instead of following an intuitive flow.	Major	Introduce a step-by-step guide with visual cues to assist users through the publishing process.
7. Flexibility and efficiency of use	No shortcuts or autofill for frequent sellers uploading similar product listings.	Cosmetic	Add a “Duplicate Post” or “Use Previous Template” option for quicker uploads.
8. Aesthetic and minimalist design	The home screen has slightly crowded sections with too many featured buttons.	Minor	Reduce the number of primary actions or group them under collapsible menus for a cleaner look.
9. Help users	When login fails, the	Major	Display specific

recognize, diagnose, and recover from errors	error message is vague ("Something went wrong").		messages like "Incorrect password. Please try again."
10. Help and documentation	There is no dedicated help or FAQ section for first-time users.	Minor	Include a "Help Center" accessible from the menu with FAQs and beginner tutorials.

6.2 Evaluation Based on Graphic Design Principles

Table 3: Graphic Design evaluation

Design Principle	Problem	Severity	Proposed Solution
Alignment	Some tutorial cards appear slightly misaligned when viewed on smaller screens.	Cosmetic	Use a responsive grid layout that ensures consistent alignment across devices.
Contrast	Text over light-colored backgrounds (e.g., white over pastel yellow) is hard to	Major	Increase text contrast by using darker font colors or semi-transparent

	read.		overlays.
Proximity	Buttons for “Edit” and “Delete” on the product detail page are too close together, increasing chances of misclicks.	Minor	Add padding or spacing between buttons to improve click accuracy.
Balance	The landing page feels visually heavy on the top due to large banners and minimal content below.	Minor	Reorganize layout to distribute content evenly—move some featured content lower or use carousels.
Hierarchy	In the “Learn” section, article headings and body text have similar font sizes, making them hard to differentiate.	Minor	Increase heading font size and apply bold styling to improve content hierarchy.

By checking our EcoTreasure app design, we found some small and a few big problems with how it looks and works. These problems showed us what to fix to make the app easier to use and more attractive. This will help us improve the design so users can enjoy the app more and use it without confusion.

Chapter 7: Product Development

7.1 Model Train (ResNet, EfficientNet, MobileNetV2, DenseNet)

In order to build the basic image classification engine of EcoTreasure: Turn Waste into Wonder!, We trained several deep learning models on the "Recyclable and Household Waste Classification" dataset available on Kaggle. The dataset consists of a wide range of labeled images for recyclable and household waste classes, which is best applicable to our sustainability app.

Model Selection

We evaluated the performance of four well-established convolutional neural network (CNN) architectures:

- ResNet (Residual Networks)
- EfficientNet
- MobileNetV2
- DenseNet

These models were selected based on their balance between classification accuracy and computational efficiency, key considerations for integration into a real-world, user-facing application.

Training Process

- All models were trained with the following settings:
- Maximum Epochs: 30
- Early Stopping: Enabled to avoid overfitting
- Loss Function: Categorical Cross-Entropy
- Optimizer: Adam
- Evaluation Metric: Accuracy (training and validation)

The training data was augmented to improve model generalization, and validation loss was closely monitored to determine optimal stopping points.

Table 4: Training Results

Model	Accuracy
ResNet	94.43%
EfficientNet	95.62%
MobileNetV2	95.84%
DenseNet	96.07%

Analysis

- EfficientNet recorded the lowest validation loss, indicative of strong generalization performance.
- DenseNet recorded the highest training accuracy and lowest validation loss.
- All the models were converging rapidly, with early stopping activated well below the 30-epoch mark.
- MobileNetV2 and ResNet performed reasonably as well, offering slightly lower complexity and inference time.

This comparative model training allowed us to choose the best-suited architectures for deployment in EcoTreasure, compromising between accuracy, efficiency, and scalability.

7.2 Development Plan

In this phase of the project, we have worked solely on the UI/UX and created a high-fidelity prototype of the mobile app EcoTreasure using Figma. The interface of the application has been crafted with user experience in consideration and there has been no backend system, machine learning, or technical implementation that has been embraced as of yet. However, this chapter provides the future development roadmap to develop EcoTreasure as a complete and working product.

Going forward, we intend to incorporate the following technologies and features into our prototype to bring our product to life:

1. Planned Technology Stack

- **Frontend:**

  We aim to utilize Flutter, Google's robust and feature-packed open-source UI toolkit, to create cross-platform Android and iOS applications from a shared codebase.

- **Backend:**

  The backend can be created with Node.js and Express.js for server-side development and API creation.

- **Database**

  For storing user data, project data, and shared data, we are looking at Firebase Firestore or MongoDB Atlas.

- **Authentication:**

  We intend to integrate Firebase Authentication to give users the option to log in using email/password or third-party services such as Google or Facebook.

- **Cloud Storage:**

  To store videos and images uploaded by users, Firebase Storage will be utilized as it is scalable and readily integrates with other Firebase services.

2. Scheduled Machine Learning Functions (Optional/Advanced)

a. Additional Features (Planned)

- **Chat and Messaging:**

  Allow users to communicate, team up, or exchange resources through in-app messaging.

- **Location-Based Services:**

  Help users find nearby recycling facilities, eco-stores, or donation drop-off locations.

- **Gamification:**

  Incorporate features such as achievements, eco-points, and badges to encourage users to contribute continuously and responsibly.

b. Development Schedule (Provisional)

- Phase 1: Frontend and Backend Development (2–3 months)

- Phase 2: Feature Integration and Testing (1–2 months)
- Phase 3: Optional ML Integration and Final Review (2 months)
- Phase 4: Release on Play Store / App Store

Although we have not yet implemented the final product, this design project has established a good foundation for the development to follow. The next immediate task is to develop the application incrementally with the proposed technologies and features while closely monitoring sustainability and the user experience.

Chapter 8: Conclusion

8.1 Conclusion

The EcoTreasure project was conceived to encourage sustainability with innovative upcycling and community-powered interaction. With a user-centered design approach and features like DIY tutorials, marketplace, and green content, the prototype can meet the needs of users seeking to minimize wastage and become greener. Through research, prototyping, and interface testing, we established a solid design framework that echoes functional objectives as well as visual principles. The project illustrates the way good design can assist people in taking small yet significant strides towards a more sustainable future.

8.2 Learnings

In the course of developing the EcoTreasure prototype, we gained valuable insights into user needs, sustainable behavior patterns, and the importance of intuitive design. From early user research to interface testing, we found that users value simplicity, graphical readability, and coaching in the practice of eco-friendly actions. We learned, as well, how important it is to map features to users' existing habits and motivations to encourage frequent use of the application.

Moreover, working with tools like Figma helped us hone our design skills and learn how to apply UI/UX principles in real-life scenarios. The iterative design process helped us address issues methodically and creatively.

8.3 Limitations

Though we could somehow provide the UI design and prototype, there were constraints to our project:

- We have not done the actual working application or any machine learning models because of time and technical limitations.
- Our user research was mostly confined to interviews and questionnaires; more extensive and varied user testing might provide deeper insights.
- The high-fidelity prototype also requires some polishing in terms of accessibility and responsiveness across various devices.

8.4 Future Plan

In the future, we aim to create a fully functional version of the EcoTreasure app. This could involve:

- Feature Development: Implementing core features like user accounts, DIY tutorials, the eco-points scheme, and community building tools.
- Machine Learning Integration: Integrate a smart recommendation system that provides upcycling suggestions according to what users possess.
- Beta Testing: Releasing a beta to a small group of users to obtain feedback and make design refinements.
- Accessibility Enhancements: Making the app compatible with various devices and accessible to individuals with disabilities.
- Scalability and Impact: Scale the app to collaborate with local eco-organizations and incorporate real-life sustainable campaigns.

This project has set a good foundation, and we are excited to see the potential that EcoTreasure can bring to make sustainable living more attainable and enjoyable.

Chapter 9: Appendices

9.1 Acknowledgment

We would also like to extend our heartfelt gratitude to our supervisor, Yasin Sazid, Lecturer, Department of Computer Science & Engineering, East West University, for his motivating guidance, recommendations, and inspiration throughout the project. His inputs regarding Human-Computer Interaction and practicing green design helped to direct the EcoTreasure prototype. We also thank all the volunteers who took their time and shared their inputs in our interview and survey. Their contribution was invaluable in identifying user needs and determining the functionalities of our app.

9.2 Working with the Raw Data: Interview Answers (Summary)

- **Participant 1 – Arpita (27, Graphic Designer)**
  - ➔ Habits: Recycles, upcycles clothing, and composts.
  - ➔ Challenges: Cost and time of eco-friendly alternatives.
  - ➔ App Expectations: Visual tutorials, daily reminders, social sharing.
  - ➔ Visuals: Very necessary; video/image preference.

- **Participant 2 – Momota (41, School Teacher)**
  - ➔ Habits: Cycles, gardens, reuses jars, recycles.
  - ➔ Challenges: Few eco-products available, no digital tools.
  - ➔ App Expectations: Group challenges, progress tracking, kid-friendly ideas.
  - ➔ Visuals: Handy for family/kids projects.

- **Participant 3 – Tasin (23, University Student)**
  - ➔ Habits: Upcycles, thrifts, low-waste living.
  - ➔ Challenges: Peer apathy, small space/means.
  - ➔ App Expectations: Social sharing, gamification, bite-sized visual content.
  - ➔ Visuals: Very important; prefers step-by-step videos or reels.

9.3 Raw Data: Online Survey (Summary)

Participants: 30 (mostly university students, ages 18–25)

Key Insights:

- 75% practice some form of sustainability
- 65% don't know how to start upcycling
- 80% want tutorial-based content
- 70% open to using a sustainability app
- **Most requested features**: Visual guides, gamification, and community sharing

9.4 Resources

- **Google Survey Form Link:** https://forms.gle/zjstf3iaoBmbKf758

- **EcoTreasure User Insight Survey (Responses) Link:** https://docs.google.com/spreadsheets/d/1Ljn2daTiYqmTXAXpeAlnXpA3vNm7N6V9S17Fi5U1LCc/edit?gid=1720383586#gid=1720383586

- **Interview Document Link:** https://docs.google.com/document/d/1JZazTWs6PPD_cgTDPBQQedx2STYAH0IzZs3tc-u7OhQ/edit?usp=sharing

- **Figma Prototype Link**:

  https://www.figma.com/design/x5NebOQaO0QSu7oUbAnAqq/EcoTreasure-Main-app?m=auto&t=m5keYw0yWxJbhdGS-6

- **Dataset Link:** https://www.kaggle.com/datasets/alistairking/recyclable-and-household-waste-classification/code
- **Model Train Code Link:** https://github.com/TithiP23/CSE-428-1--2021-2-60-057